

Alessandro Cascioli

ENERGY ENGINEER · PHD

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Work Experience

Saldi e Partners

Perugia, Italy

ESG, DIGITALIZATION AND MANAGEMENT SYSTEMS CONSULTANT

Jun. 2025 - Present

- Provide strategic ESG consulting focused on sustainability and the digitalization of business processes.
- Lead sustainability impact assessments, identifying key risks and opportunities for business growth.
- Support organizations in adopting ISO 27001 Information Security Management Systems.
- Implement and develop ISO/IEC 42001 Artificial Intelligence Management Systems.
- Perform risk analysis, define operating procedures, and optimize digital document workflows.
- Train and raise staff awareness on regulatory requirements and digital management tools.
- Support internal and external audits, ensuring regulatory compliance and continuous improvement.

KPMG

Perugia, Italy

INDUSTRY 4.0 AND 5.0 TRANSITION CONSULTANT AND APPRAISER

Sep. 2024 - Present

- Deliver technical and strategic consulting for innovation and digital transformation projects within the Industry 4.0 and 5.0 transition.
- Prepare sworn technical reports to secure tax incentives and benefits.
- Analyze production processes and identify solutions for automation and systems integration.
- Assess digital technologies, IoT, AI, and advanced control systems.
- Support companies in implementing sustainability and energy efficiency solutions.
- Verify compliance with regulatory requirements and prepare technical documentation.
- Assist during audits and inspections conducted by the relevant authorities.

Syntex GreenTech

Perugia, Italy

SUSTAINABILITY CONSULTANT

Oct. 2023 - Present

- Develop innovative ESG strategies that improve sustainability and reduce financial risks through data-driven decision-making.
- Conduct sustainability impact assessments, identifying key risks and opportunities for business growth.
- Prepare sustainability statements in line with the GRI and EFRAG frameworks.
- Develop a platform that accelerates information gathering and KPI monitoring.
- Support clients across footwear, pharmaceutical, logistics, transportation, apparel, energy, and food retail sectors.

I.T.S. Umbria Academy

Perugia, Italy

LECTURER

Oct. 2023 - Present

- Lecturer for the training unit "Project Management" for 18 hours within the course "Energy Efficiency".

Free University of Bozen-Bolzano

Bolzano, Italy

POSTDOCTORAL RESEARCHER AND TEACHING ASSISTANT

Nov. 2022 - Sep. 2025

- RFD project - Recovery of strategic materials from sewage sludge (acronym RFD).
- FRONTSHIP project - A frontrunner approach for the transition to a circular and resilient future: deployment of systemic solutions with the support of local clusters and the development of regional community-based innovation schemes.
- Carry out techno-economic analysis to optimize process conditions and reduce costs.
- Develop simulations and models to enhance the understanding of thermochemical processes.
- Apply data analysis techniques to experimental datasets, including peak deconvolution and curve fitting.
- Develop a machine learning model to predict biocrude oil yield.
- Teaching assistant for the course "Energetics" in the B.Sc. in Industrial and Mechanical Engineering.
- Teaching assistant for the course "Power Production, CHP and District Heating Systems" in the M.Sc. in Energy Engineering.
- Support the supervision of PhD and Master's students.

SGS

Italy

ISO 27001:2013/2022 AND ISO 42001:2023 AUDITOR

Aug. 2021 - Present

- Lead auditor training for ISO/IEC 27001:2013 Information Security Management Systems.
- ISO/IEC 27001:2022 Information security, cybersecurity, and privacy protection.
- ISO/IEC 42001:2023 Artificial Intelligence Management Systems.
- Organize webinars and training courses on ISO 42001:2023.

SGS

Italy

CONSULTANT

Oct. 2018 - Oct. 2019

- Deliver dealer training for stock audits on assignment from a certifying organization for a leading heavy machinery manufacturer.
- Conduct audits in Italy.

Qualitas S.r.l.

CONSULTANT AND AUDITOR

- Conduct ISO/IEC 27001 Information Security Management internal audits for a consulting company.

Perugia, Italy

Jul. 2018 - Present

SGS

AUDITOR

- Conduct second-party compound audits on assignment from a certifying organization for a leading automobile manufacturer.
- Carry out audits in Italy and abroad (Austria, Germany, Finland, Poland, Portugal, Sweden).

Europe

Apr. 2017 - Oct. 2019

Baruffaldi S.r.l.

CONSULTANT

- Manage CE homologation processes for motorcycle helmets, goggles, and gloves.
- Handle relationships with suppliers, mainly outside Italy (China and Pakistan).

Perugia, Italy

Oct. 2016 - Oct. 2019

Skills

Programming	SQL, Python, JavaScript, TypeScript, C, Bash, Dart, LaTeX, Matlab
Scientific computing	NumPy, SciPy, Sympy, Pandas, LMFIT
Machine learning	TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost
Data visualization	Tableau, Matplotlib, Seaborn, Plotly, Altair, Sigmaplot
Dashboards and app development	Streamlit, Dash, Django, Flask, React, React Native, NextJS, AstroJS, Flutter, PySide, TKinter
Database management	PostgreSQL, MySQL, SQLite, MongoDB, SuperbaseDB

Certificates

2025	ISO/IEC 9001:2015 Quality Management Systems Auditor/Lead Auditor	Dimitto AG
		IDCERT
2025	IDCERT DPO/RPD	International Digital Certification
2025	ISO 14001:2015 Environmental Management Systems Auditor/Lead Auditor	Dimitto AG
2024	ISO/IEC 42001:2023 Artificial Intelligence Management Systems Auditor/Lead Auditor	Dimitto AG
2023	ESG-SDGs:2022© Assessment Course for Auditors/Lead Auditors and Consultants	CERTIFICATION Srl
2023	ISO/IEC 27001:2022 Transition Course	SGS
2023	DeepLearning.AI TensorFlow Developer Specialization	Coursera
2020	State examination for the qualification to practice as an Energy and Nuclear Engineer	University of Trento
2019	ISO/IEC 27001:2013 Information Security Management Systems Auditor/Lead Auditor Training Course	SGS

Education

Free University of Bozen-Bolzano

Bozen, Italy

PHD IN SUSTAINABLE ENERGY AND TECHNOLOGIES

Oct. 2019 - Oct. 2022

- Thesis: Hydrothermal liquefaction of biomass and biocrude upgrading: fundamentals, experimental analysis and modelling.
- Spent 5 months as a visiting PhD in the Advanced Biofuels Lab of Aalborg University, Denmark.
- Developed and commissioned a pilot plant for the production of biocrude oil to be upgraded to SAF (Sustainable Aviation Fuels).
- Process modelling through optimization algorithms to find thermodynamic equilibrium.
- Performed techno-economic analysis to optimize process conditions and reduce costs.
- Support in the supervision of master students.
- Preparation of technical reports, visualizations, and presentations.

Free University of Bozen-Bolzano

Bozen, Italy

M.Sc. IN ENERGY ENGINEERING

Sep. 2017 - Oct. 2019

- Thesis: Valorization of residues from parapharmaceutical and herbal medicine production: a case study.

University of Perugia

Perugia, Italy

B.E. IN MECHANICAL ENGINEERING

Sep. 2013 - Apr. 2017

- Erasmus + of 6 months at the FAU Erlangen-Nürnberg, Germany.

Languages

Italian	Native or bilingual proficiency, C2
English	Professional working proficiency, C1
German	Independent user, B2

Appendix

Publications

78th national congress ATI (2023)

V. BENEDETTI, A. CASCIOLI, S. FASOLI, A. AREF, F. PATUZZI, M. BARATIERI

2023

- Continuous Hydrothermal Liquefaction of microalgae for biocrude oil production

Bioresource Technology

V. BENEDETTI, A. CASCIOLI, M. PECCHI, M. BARATIERI

2024

- Fast screening of catalyst performances in hydrothermal processes for biofuel production using differential scanning calorimetry
- DOI: <https://doi.org/10.1016/j.biortech.2024.130934>

Fuel Processing Technology

M. PECCHI, A. CASCIOLI, A. R. MAAG, J. L. GOLDFARB, M. BARATIERI

2022

- Uncovering the transition between hydrothermal carbonization and liquefaction using differential scanning calorimetry
- DOI: <https://doi.org/10.1016/j.fuproc.2022.107349>

Fuel

A. CASCIOLI, M. BARATIERI

2021

- Enhanced thermodynamic modelling for hydrothermal liquefaction
- DOI: <https://doi.org/10.1016/j.fuel.2021.120796>

Conferences and Seminars

WasteEng22 - The 9th International Conference on Engineering for Waste and Biomass Valorisation

[Copenhagen, Denmark](#)

ORAL

2022-06-27 - 2022-06-30

Legitimation of differential scanning calorimetry for the study of hydrothermal carbonization and liquefaction.

EUBCE 2022 - 30th European Biomass Conference and Exhibition

[Online](#)

ORAL

2022-05-09 - 2022-05-12

A Newly Designed Pilot-Scale Continuous Reactor for Hydrothermal Liquefaction: Preliminary Results and Validation of an Enhanced Thermodynamic Model.

WasteEng21 - The 8th International Conference on Engineering for Waste and Biomass Valorisation

[Online](#)

ORAL

2021-05-31 - 2021-06-04

Valorisation of lignocellulosics residues from herbal medicine production.

EUBCE 2021 - 29th European Biomass Conference and Exhibition

[Online](#)

ORAL

2021-04-26 - 2021-04-29

Hydrothermal carbonization of lignocellulosic biomass from para-pharmaceutical and herbal medicine production.

EUBCE 2020 - 28th European Biomass Conference and Exhibition

[Online](#)

POSTER

2020-07-06 - 2020-07-09

Exploitation of lignocellulosic biomass from para-pharmaceutical and herbal medicine production.

Honors & Awards

INTERNATIONAL AWARDS

2022 **Springer Nature's Best Oral Flash Award**, WasteEng2022 Conference

[Copenhagen, Denmark](#)

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